

Baoshan Song

Curriculum Vitae

The Hong Kong Polytechnic University PQ502

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Education

- 2023–present **PhD Degree - The Hong Kong Polytechnic University, Hong Kong.**
Thesis Title: Certifiable Multi-Sensor Integrated Navigation Based on Factor Graph Optimization for Autonomous Ground Vehicles
(Supervised by **Dr. Li-Ta Hsu** and **Dr. Weisong Wen**)
Status: In the writing phase
- 2025 **Visiting PhD Student - McMaster University, Canada.**
Project Title: Certifiable Multi-Camera Hand-Eye Calibration
(Supervised by **Dr. Matthew Giamou**)
Status: Finished
- 2020–2023 **M.Sc. Engineering, Wuhan University, China.**
Thesis Title: Multi-Robot Decentralized Mapping Based on Manifold Optimization
(Supervised by **Dr. Xingxing Li** and **Dr. Yun Wu**)
Graduation grade: very-good
- 2016–2020 **B.C.s Engineering, Wuhan University, China.**
Graduation project: Real-time GNSS/INS loosely-coupled localization system
(Supervised by **Dr. Xingxing Li**)
Graduation grade: Distinguished

Professional Interests

Working in the research fields of factor graph optimization (FGO), convex optimization and their application on simultaneous localization and mapping (SLAM).

Publications

Journals

- **B. Song**, W. Wen, Q. Zhang, B. Xu and L. Hsu. Certifiably Optimal Doppler Positioning With Opportunistic LEO satellites. *IEEE Transactions on Aerospace and Electronic Systems* (Under Review).
- **B. Song**, X. Xia, P. Yan, Y. Zhong, W. Wen and L. Hsu. Online IMU-Odometer Calibration using GNSS Measurements for Unmanned Ground Vehicle Localization. *IEEE Transactions on Intelligent Transportation Systems* (Major Revision).
- P Yan, **B. Song**, X Xia, W Wen and LT Hsu. An Efficient Detector for Faulty GNSS Measurements Detection With Non-Gaussian Noises. *NAVIGATION: Journal of the Institute of Navigation* (Under Review).
- **B. Song**, R. Xu, Z. Zhan and L. Hsu. Degeneration of Sliding-Window Factor Graph Optimization into Iterated Extended Kalman Filtering. *Nature Portfolio journal Wireless Technology* 2026.
- **B. Song**, M. Giamou, P. Yan, C. Xia and L. Hsu. Certifiable Alignment of GNSS and Local Frames via Lagrangian Duality. *IEEE Robotics and Automation Letters* 2026.
- X. Li, **B. Song**, et.al. Consistent Localization for Autonomous Robots with Inter-vehicle GNSS Information Fusion. *IEEE communications letters* 2022.
- X. Li, Z. Qin, Z. Shen, X. Li, Y. Zhou and **B. Song**, A High-Precision Vehicle Navigation System Based on Tightly Coupled PPP-RTK/INS/Odometer Integration. *IEEE Transactions on Intelligent Transportation Systems* 2022
- X. Li, Y. Zhou, Z. Shen, **B. Song** and S. Li. Using a Moving Antenna to Improve GNSS/INS Integration Performance Under Low-Dynamic Scenarios. *IEEE Transactions on Intelligent Transportation Systems* 2022

Preprint

- **B. Song**, P Yan, X Xia, Y Zhong, W. Wen and L. Hsu. Two stage GNSS outlier detection for factor graph optimization based GNSS-RTK/INS/odometer fusion. *arXiv preprint arXiv:2510.00524*.
- Z. Li, P. Yan, **B. Song** and L. Hsu. Improved GNSS Positioning in Urban Environments Using a Logistic Error Model. *arXiv preprint arXiv:2603.16420*.

Conferences

- **B. Song**, F. Huang, L. Hsu. A Decentralized LiDAR-SLAM System with Certifiably Optimal Pose Graph Optimization. *IEEE ICRA 2026 workshop*.
- Y. Zhang, Y. Wang, **B. Song** and W. Wen. Integrity-Monitored Deep Reinforcement Learning for Safe and Robust Autonomous Navigation. *IEEE ITSC 2025*.
- **B. Song**, W. Wen, L. Hsu. Certifiably Optimal Satellite Orbit Determination Based on Doppler Measurements for Low-Earth-Orbit Satellite. *IEEE/ION PLANS 2025*.
- **B. Song**, W. Wen, L. Hsu. Tightly coupled Low-cost GNSS-RTK/INS/Odometer Integration Via Factor Graph Optimization Aided by GNSS Outlier Mitigation in Urban Canyons. *European Navigation Conference (ENC) 2024*.

Patents

- **B. Song**, et.al. An integrated positioning method based on ArduRover unmanned vehicle. *Chinese Patent ZL202110539154.8, 2021.*

Work Experience

- 2025-2026 **Teaching Assistant**, *Postgraduate Course: Robust Control Technology In Low-Altitude Aerial Vehicle.*
Task: Teach ORB-SLAM3 coding tutorial and 3D Gaussian Splatting course.
- 2023-2024 **Teaching Assistant**, *Undergraduate Course: Dynamic systems and control.*
Task: Design teaching materials and teach the PID control experiments using robot operation system (ROS).
- 2020-2021 **Part-time Engineer**, *Hangzhou Shenhao Technology Co..*
Task: Develop a simulated rail track irregularity measuring system using tactical IMU.

Supervision

- B.Sc **Yuxuan Tan (WHU,2021)**, *PPP/INS/Vision tightly coupled integration navigation in complex environments.*
- M.Sc. **Minghe Han (PolyU,2026)**, *Certifiably optimal semantic LiDAR mapping.*

Skills and Activities

- Programming C/C++, MATLAB, Python, Android
- Third-party ROS, Qt, OpenCV, Latex, SOLIDWORKS
Library
- Operating Windows, Mac, Linux
System

Languages

- Cantonese Native Speaker
- Mandarin Mandarin (Secondary level-A)
- English English (IELTS)(PhD in English, PhD dissertations written in English, publications and presentations are in English)

Honors and Awards

- 2020 Wuhan University Outstanding undergraduate graduate
- 2019 Wuhan University Research Postgraduate Recommendation (*the only one*)
- 2019 First Prize in Wuhan university surveying competition
- 2018-2019 Wuhan University Outstanding Student
- 2017-2018 National Encouragement Scholarship
- 2017 Wuhan University Merit Student

Services

- Reviewer NAVIGATION, TAES, TVT, TIE, ITSC, IV